



MATHS OLYMPIAD CHALLENGE: INTRODUCTION TO MATHEMATICAL REASONING & PATTERN SPOTTING



WELCOME TO THE DIWALI MELA PUZZLE!

Kabir and Diya are at the Diwali Mela when they discover a magical puzzle box! The box is locked by a secret pattern code. Colorful stalls, toran decorations, and festive lights surround them as they try to figure out the mystery.

**Help Kabir & Diya unlock
the magic box by spotting
the right pattern!**



WHAT IS A PATTERN?

Look at this toran decoration:
Mango Leaf → Marigold Flower → Mango Leaf → Marigold Flower → ?
A Pattern is when shapes, colours, or numbers repeat in a rule!
Can you spot what comes next?

Pattern: Things that repeat in a rule!

Sequence & Rule: What changes?
What stays the same?





MORE PATTERNS AROUND US!

Numbers make patterns too! Look at the hopscotch — 3, 6, 9, 12... Can you see the rule? We skip count by 3s! Even ₹5 and ₹10 notes follow a pattern. What number comes next?

Hopscotch: 3, 6, 9, 12,
__?
Skip count by 3s!

₹10 notes: 10, 20,
30, __?
What comes next?

₹5 notes: 5, 10, 15,
__?
Spot the pattern!

Fill the empty box —
use the Sequence
Rule!

OLYMPIAD PUZZLE WALKTHROUGH #1

Visual Pattern Rotation

Look at the triangles below. Each one is rotated in a sequence.

- ▲ What do I see?
- 🔄 What changes?
- ✓ What stays the same?



**Find the rule, then
predict the next
triangle!**



OLYMPIAD PUZZLE WALKTHROUGH #2

The Number Wheel Mystery ^{1 2}_{3 4}

Step 1: What do I see? A wheel with numbers.
Step 2: What is the Rule? Spot the **Sequence**!
Step 3: Follow the rule to find the missing number.
Step 4: Verify — does it fit the pattern?



Key Words: Sequence & Rule — use them to solve any number puzzle!



THE CHALLENGE ZONE – QUESTION 1 ★

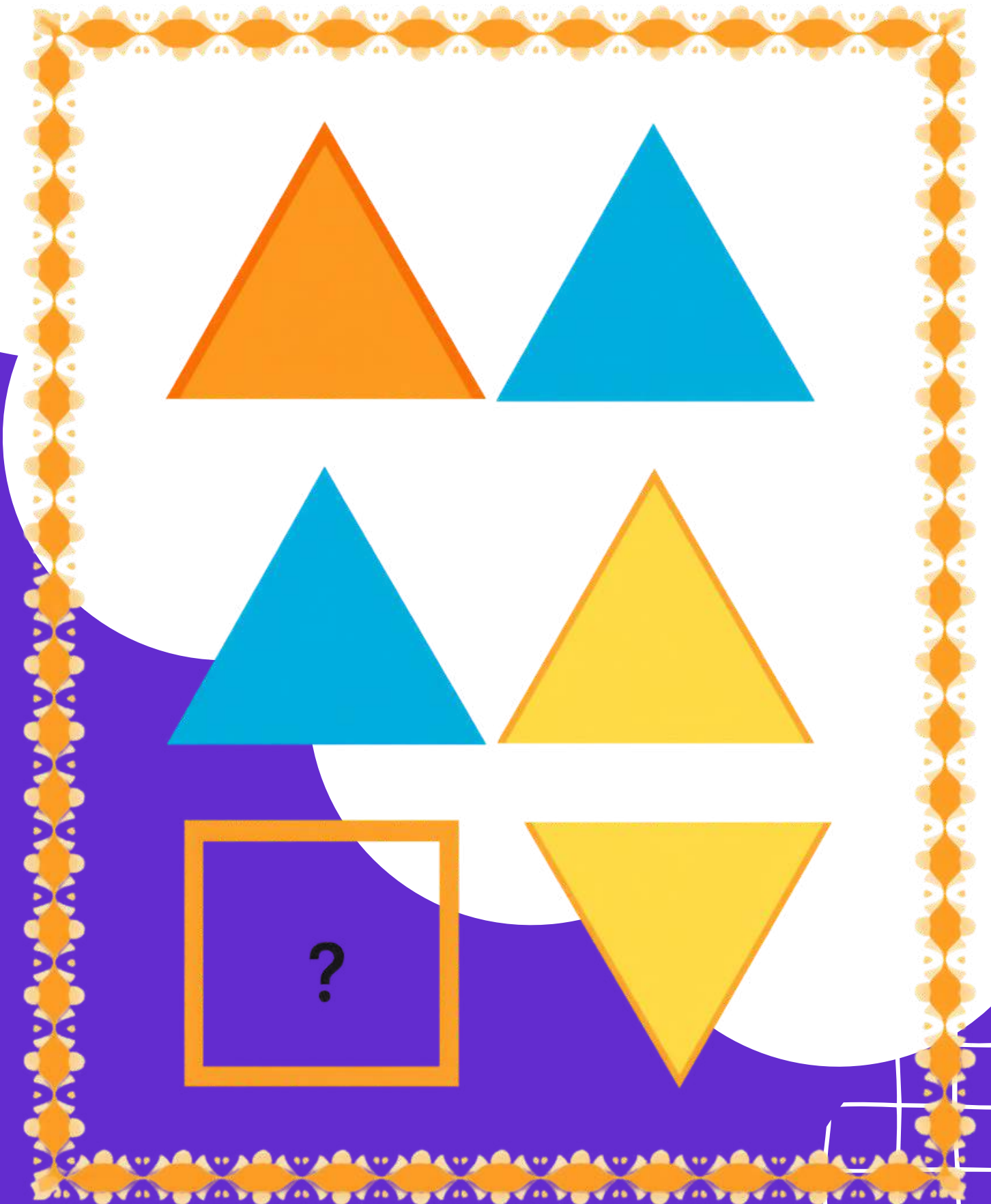
Look at the Rangoli boundary pattern below:

▲ Triangle UP → ► Triangle RIGHT → ▼ Triangle DOWN → ?

What comes NEXT in the pattern?

- (A) ▲ Triangle UP
- (B) ◄ Triangle LEFT
- (C) ► Triangle RIGHT
- (D) ▼ Triangle DOWN

💡 **Hint: Look carefully — what direction comes after DOWN in the repeating sequence? Think about what changes each time!**



THE CHALLENGE ZONE – Q2

Hopscotch Skip Counting!
Look at the numbers on the hopscotch squares:

$4 \rightarrow 8 \rightarrow 12 \rightarrow ?$

A) 14 B) 15 C) 16 D) 18

Drag the correct number to the empty square!

Hint: What number are we adding each time? Count by the same number! ✨



THE CHALLENGE ZONE!

Question 3 (Intermediate) 🌟

Tarazu Puzzle: Kabir is weighing fruits at the Diwali Mela!

Use the balance scale clues to find the answer.

2 Apples = 1 Mango | 2 Mangoes = 1 Papaya

? How many Apples balance 1 Papaya?

A) 2 Apples

B) 4 Apples ✓

C) 6 Apples

D) 8 Apples



THE CHALLENGE ZONE – QUESTION 4

Advanced Olympiad Level

Look at the Warli painting grid below. Each row and column follows a hidden pattern of shapes. Find the missing shape to complete the sequence!

A) Triangle B) Circle C) Square D) Diamond



Hint: What changes in each row? What stays the same in each column?



COMMON PITFALLS – MOJO'S MISTAKES!

It's okay to make mistakes! Let's learn to spot them and get better. Watch out for these tricky traps when solving pattern puzzles!

Keywords: Symmetry | Sequence

Forgetting to count the outer triangle in a kite design – always check the edges!

Miscounting shapes in a sequence – slow down and count each one carefully!



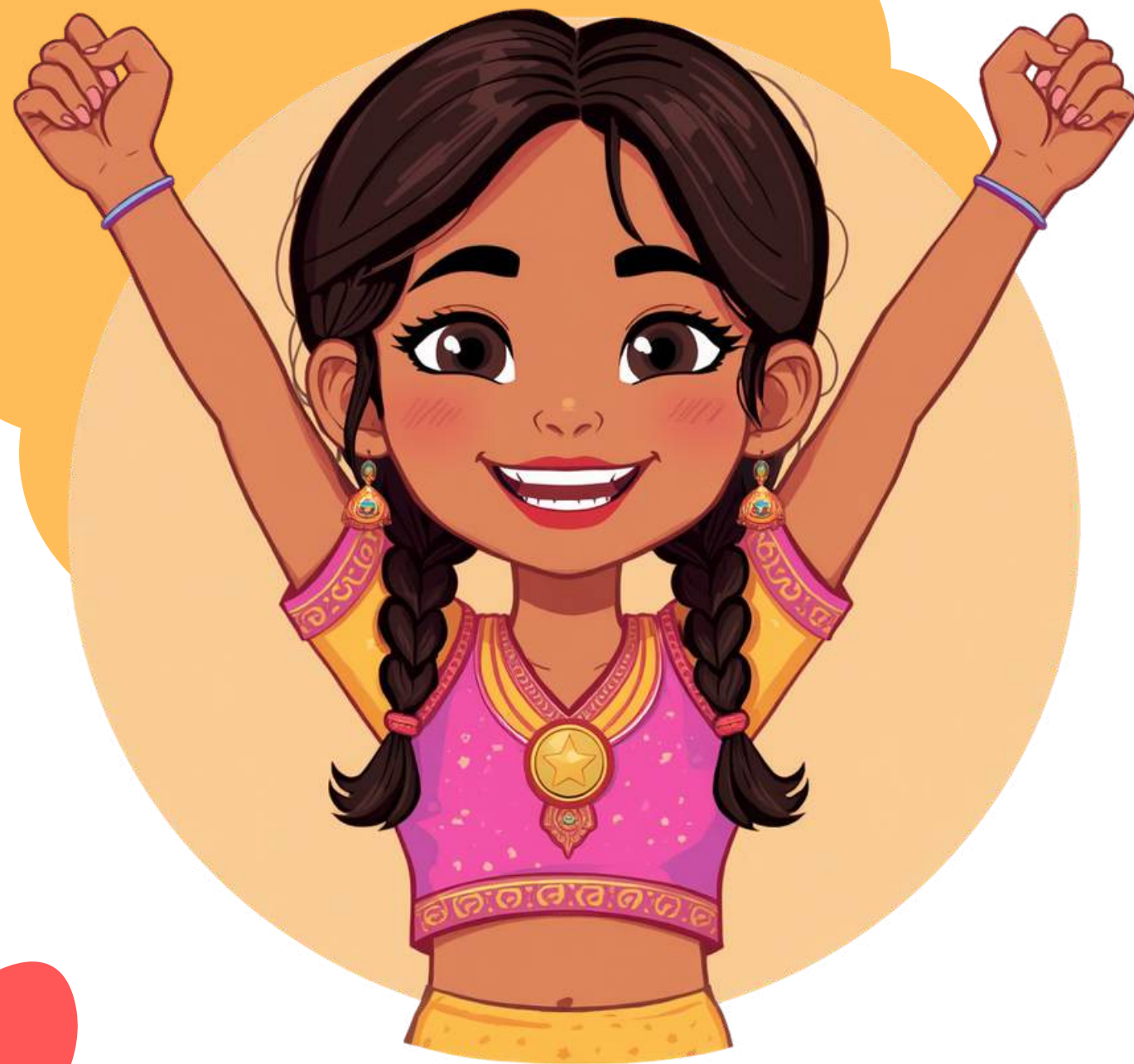
RECAP – YOUR SUPERPOWERS TODAY!

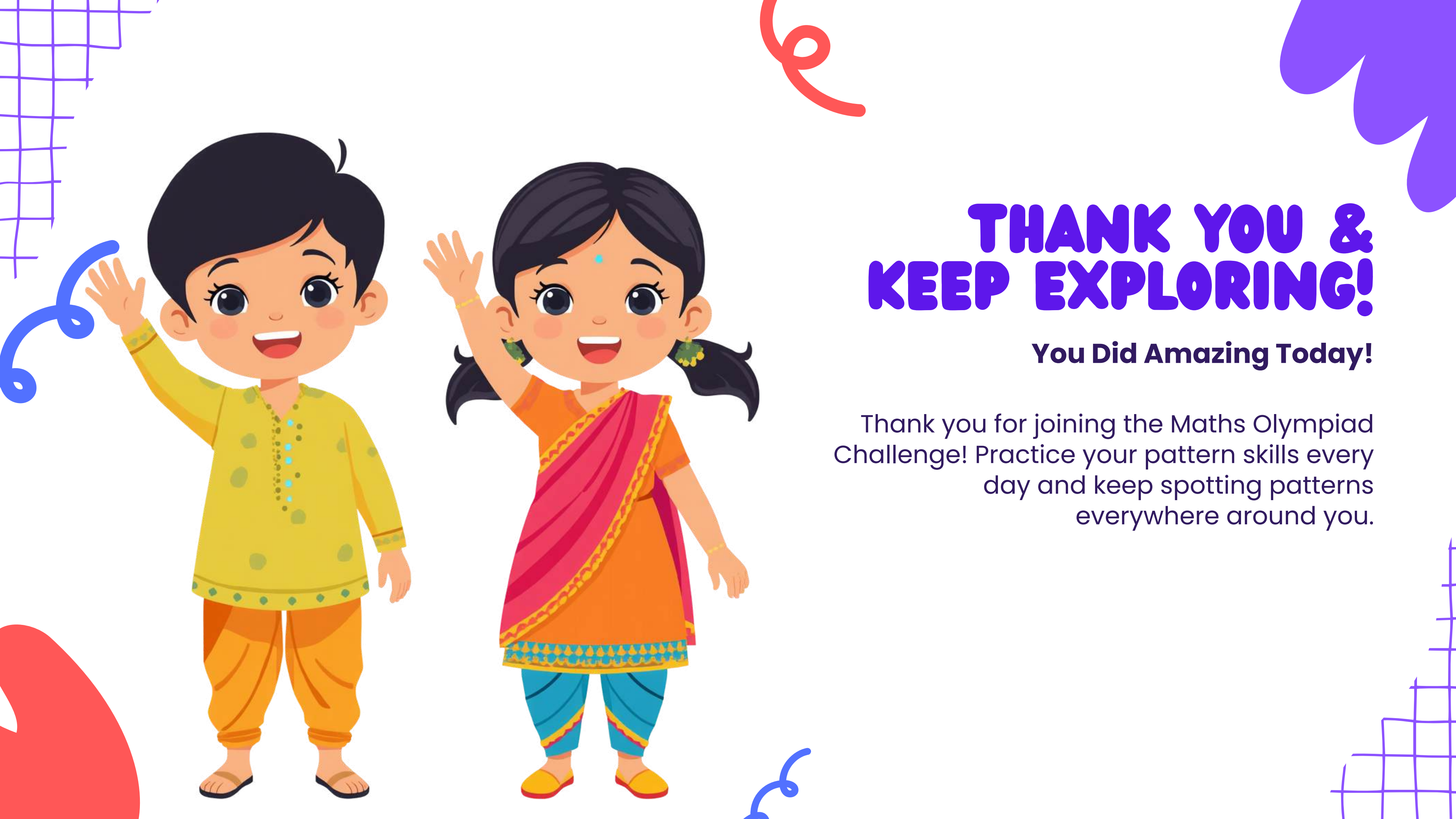
Amazing work today! You have unlocked powerful thinking skills. Here is what you discovered in this session. You are now a Pattern Detective!

Spot Patterns & Sequences around you every day!

Use logical thinking with Balance Scales (Tarazu)!

Ask smart questions: "What changes? What stays the same?"





THANK YOU & KEEP EXPLORING!

You Did Amazing Today!

Thank you for joining the Maths Olympiad Challenge! Practice your pattern skills every day and keep spotting patterns everywhere around you.

